



# Mobility System Design and Testing for GREMLINS Micro-rovers



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## Background and Research Goals

The GREMLINS project develops multi-rover teams that focused on testing locomotion strategies to handle slippery rover motion under lunar regolith in low gravity conditions. Our team decided to utilized a similar small-scale rover version with treads in the NBRF lab to specifically focus on how different mobility designs will affect performance on rough terrain with different slopes while carrying lunar material. We compared the installed rover treads with alternate rover wheel design ideas to determine which option could offer better traction and mobility on loose lunar regolith, focusing on different grouser shapes and comparing it with slippage, distance, and speed during climbs. What we defined to be a successful climb was the ratio of distance travelled to the slip of the wheels at a certain threshold, or how much more the wheel spined compared to how far the rover moved. We hope the findings will help inform the design of GREMLINS micro-rovers towards improved locomotion under lunar conditions

## Methodology

- Conducted control sandbox tests with rover treads
- Assigned grouser type design to each member and 3D printed wheel prototypes in CAD
- 6 grouser designs: Chevron, Spikes, Multi-Chevron, Straight, Slant, Curved
- Installed 4 new motors on rover to account for four wheels
- Conducted new sandbox tests with different weights & slopes for different wheels
- Measure the traction, slippage, climb success, and average speed for each configuration
- Discussed on trends relating to each motion mechanism with the data gathered

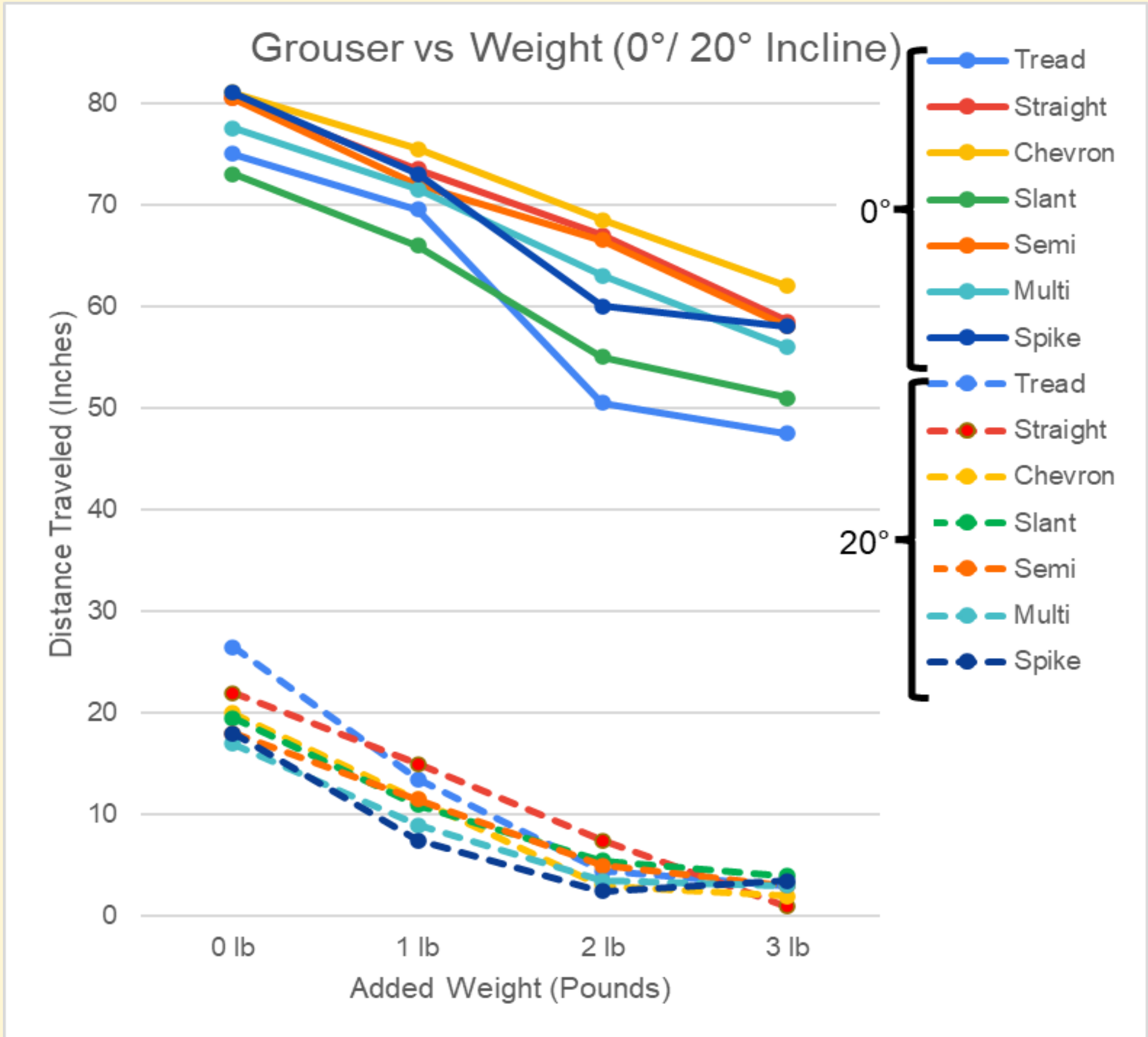


Figure 4: Different Grouser patterns effect on Rover Mobility with varying weight and incline

## Teamwork & Takeaways

- Determined problem upon initial rover analyzation in sandbox and brainstorm research ideas
- Focused on design, prototyping, testing, and data analysis
- Did iterative design: build → test → analyze → redesign
- Learned how mobility system design affects rover performance on slopes and loose terrain
- Trend analysis showed that different locomotion options provide different advantages at different variables.

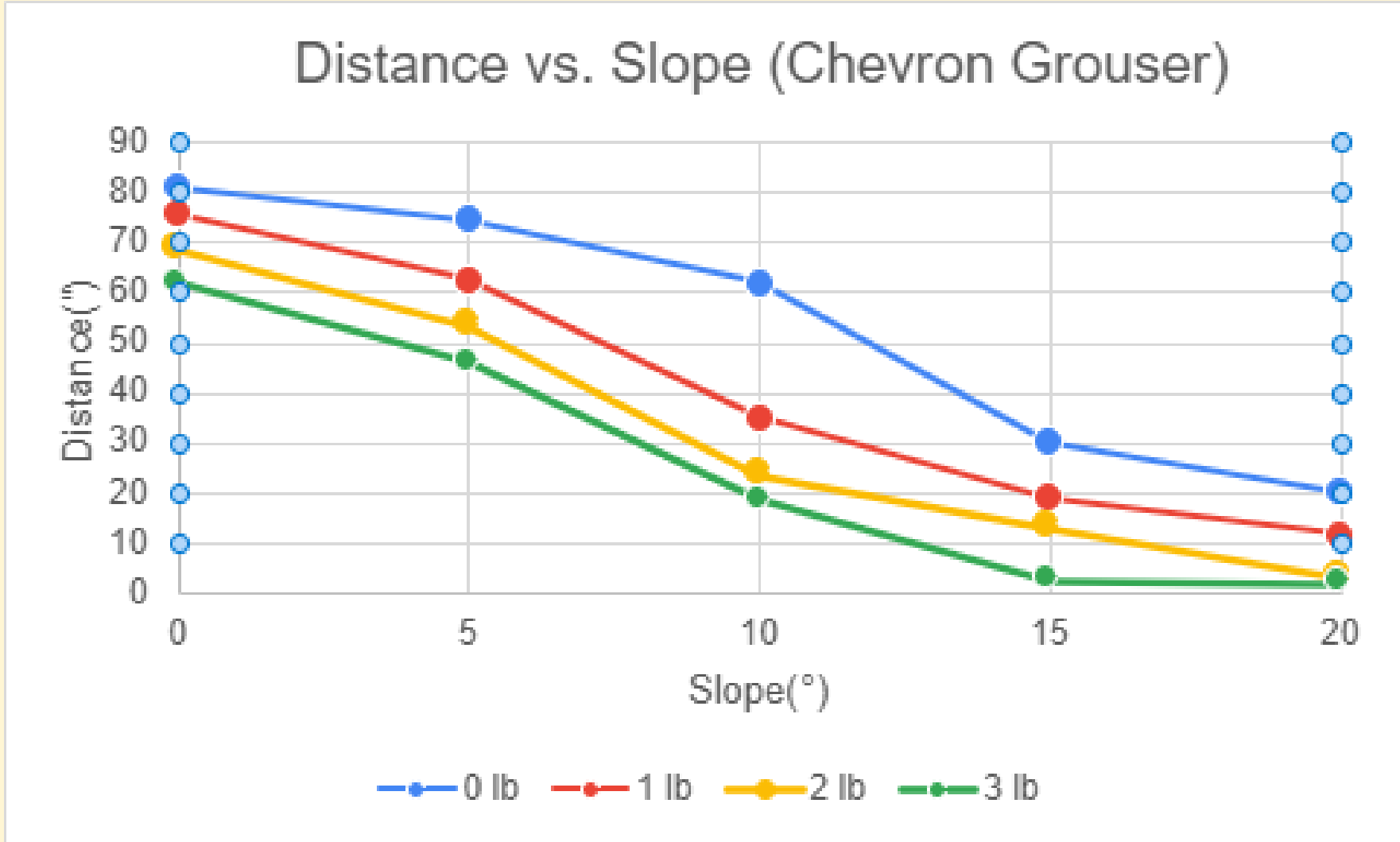


Figure 1: Rover Mobility with Chevron Grouser



Figure 2: CAD model of a Double Chevron Grouser design



Figure 3: Micro-rover on testing platform. Testing different slopes and weight

## Hypothesis / Objectives

- Wheels expected to outperform treads in overall mobility/ speed
- Hypothesis: Chevron grousers predicted to provide best traction and stability on loose regolith at multiple slopes
- Goal: Identify the optimal combination of mobility system and grouser design for GREMLINS lunar micro-rovers

## Conclusion / Outcome

- Preliminary data shows wheels perform slightly better on moderate slopes, while treads excel on looser sand
- GREMLINS rover design results can be implicated for scalable lunar construction
- Data analysis results => Our pick best grouser: Straight

Table 1: Best Grouser under different variables

|     | 0°      | 5°       | 10°      | 15°      | 20°      |
|-----|---------|----------|----------|----------|----------|
| 0lb | Spike   | Spike    | Straight | Straight | Treads   |
| 1lb | Chevron | Slant    | Straight | Treads   | Straight |
| 2lb | Chevron | Straight | Straight | Straight | Straight |
| 3lb | Chevron | Chevron  | Slant    | Slant    | Slant    |